

JUNLI GONG

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RESEARCH INTERESTS

Efficient machine learning systems—LLM inference, GPU kernels, and heterogeneous accelerators—and multimodal models that perceive, reason, and act.

EDUCATION

Northeastern University

Master of Science in Computer Science, Khoury College of Computer Sciences

Sep 2025 – Dec 2026 (expected)

Seattle, WA, USA

- **GPA:** 3.89/4.00; **Seattle Campus Scholarship** (merit-based)
- **Relevant Courses:** Machine Learning, Scalable Distributed Systems, Reinforcement Learning, Algorithms, Foundations of AI, Practical AI

Beijing Normal-Hong Kong Baptist University (BNBU)

Bachelor of Science (Honours) in Computer Science and Technology

Sep 2021 – Jun 2025

Zhuhai, China

- **Honors:** President's Honour Roll; Dean's List (4 semesters)
- **Relevant Courses:** Neural Networks and Deep Learning, Machine Learning, Data Structures and Algorithms, Operating Systems, Database Management Systems

PUBLICATIONS

* denotes equal contribution.

Peer-reviewed

- [1] Guo, Y.* , **Gong, J.***, Cai, H., Cheung, Y.-m., & Su, W. (2026). Can Segmentation Models Understand the World? Proactive Affordance Reasoning via Vision-Grounded Chain-of-Thought. *Proceedings of EMNLP 2026* (Main Conference).
- [2] Guo, Y., **Gong, J.**, Wang, W., Cai, H., Cheung, Y.-m., & Su, W. (2026). PEAM: Parametric Embodied Agent Memory for Continual Skill Learning via Internalization of Experience in Minecraft. *Findings of EMNLP 2026*.
- [3] Guo, Y., **Gong, J.**, Dong, W., Cheung, Y.-m., & Su, W. (2026). Adding Thermal Awareness to Visual Systems in Real-Time via Distilled Diffusion Models. *CVPR 2026 Workshop on World Models Meet Active Sensing (WMAS)*; extended version under review at NeurIPS 2026.
- [4] Zheng, C.* , **Gong, J.***, Guo, J., Tang, Z., & Wang, T. (2025). Enhancing Federated Learning in IoT Through Dynamic Spectral Clustering. *WASA 2025*.
- [5] Fei, C., Xia, N., Tsai, P.-W., Lu, Y., Pan, X., & **Gong, J.** (2024). An Effective Feature Selection Algorithm for Machine Learning-based Malicious Traffic Detection. *AsiaJCIS 2024*.

Under Review

- [6] Guo, Y.* , **Gong, J.***, Lu, Y., Xu, X., Cheung, Y.-m., & Su, W. (2026). Bringing Multimodal Large Language Models to Infrared-Visible Image Fusion Quality Assessment. Under review at *NeurIPS 2026*. arXiv:2605.06969.
- [7] Guo, Y., **Gong, J.**, Cheung, Y.-m., & Su, W. (2026). LumiVideo: An Intelligent Agentic System for Video Color Grading. Under review at *AAAI 2027*. arXiv:2604.02409.
- [8] Guo, Y., **Gong, J.**, Cheung, Y.-m., & Su, W. (2026). Fuse4Seg: Image Fusion for Multi-Modal Medical Segmentation via Bi-level Optimization. Under review at *AAAI 2027*. arXiv:2409.10328.

RESEARCH EXPERIENCE

Graduate Researcher

Khoury College of Computer Sciences, Northeastern University

Jan 2026 – Present

Advisor: Prof. Yifan Hu

- Investigating speculative expert pre-dispatch for Mixture-of-Experts inference: lightweight linear probes predict expert routing from pre-attention hidden states so that the expert-parallel AllToAll can overlap with attention, while a verify-and-fallback step keeps outputs identical to the serial path.
- Built a prototype in SGLang and evaluated it on DeepSeek-V2-Lite on 2×A100; currently refining the routing predictor and the evaluation methodology.

Research Assistant

Department of Computer Science, Beijing Normal-Hong Kong Baptist University (remote)

Sep 2025 – Present

Supervisor: Prof. Weifeng Su

- Responsible for the experimental work of six papers on multimodal perception and embodied AI—implementing the methods, running training and evaluation, and writing the experiments sections; two accepted at EMNLP 2026 (Main and Findings), one at a CVPR 2026 workshop, and three under review at NeurIPS 2026 and AAAI 2027 [1–3, 6–8].
- Co-first-authored SegWorld [1] (EMNLP 2026): implemented two-pass proactive affordance reasoning with vision-grounded chain-of-thought on a Sa2VA-based segmentation MLLM and evaluated it on the InstructPart intent-to-part benchmark.

- Co-first-authored FuScore [6]: built the MLLM-based infrared–visible fusion quality assessor, from zero-shot scoring to fine-tuning with soft-label distributional supervision and Thurstone-based ranking objectives, reaching state-of-the-art correlation with human preference.
- Implemented and evaluated the parametric agent-memory framework (MoE-LoRA adapters, failure–correction contrastive training) in Minecraft [2] and the distilled-diffusion real-time thermal–visible fusion module with closed-loop CARLA driving evaluation [3].

Research Intern

Jul 2025 – Aug 2025

Pazhou Lab / School of Computer Science and Engineering, South China University of Technology Supervisor: Prof. Lu Lu

- Developed a custom FP16 Tensor Core HGEMM kernel in CUDA for Transformer workloads on NVIDIA A800, combining coalesced global-memory access ($\sim 10\times$ gain over the strided-access version) with double-buffered software pipelining (+6–9%).
- Benchmarked against cuBLAS on five BERT/LLaMA-derived matrix shapes; outperformed cuBLAS by $1.33\times$ on the bandwidth-bound attention-score shape ($1024 \times 1024 \times 64$) and produced a bottleneck analysis guiding further optimization.
- Systematically evaluated shared-memory tiling on 1,000 random GEMM shapes, showing a robust $1.15\times$ mean speedup over naive GPU kernels and $100\text{--}500\times$ over a serial CPU baseline.

Undergraduate Researcher

Jan 2024 – Jun 2025

Advanced Institute of Natural Sciences, Beijing Normal University, Zhuhai Supervisor: Prof. Jianxiong Guo

- Co-first-authored a WASA 2025 paper [4] on clustered federated learning for IoT: designed a dynamic spectral-clustering scheme that groups clients by data heterogeneity to improve accuracy and convergence while preserving privacy.
- Implemented the adaptive clustering algorithms and a dynamic resource-allocation strategy in Python and evaluated scalability on large-scale simulated IoT networks.

Research Assistant

Jan 2024 – Jun 2024

Malicious Traffic Detection Project, Nanjing Normal University (remote) Supervisor: Dr. Nian Xia

- Collected and preprocessed network-traffic datasets (TII-SSRC-23, UNSW-NB15), trained SVM and decision-tree detectors with the EFS-CST feature-selection method, and wrote the data-collection and model-evaluation sections of the AsiaJCIS 2024 paper [5].

INDUSTRY EXPERIENCE

iFLYTEK Co., Ltd.

Jul 2026 – Aug 2026

AI Infra Intern Hefei, China

- Developed and debugged an automated evaluation platform for adapting speech models (ASR/TTS) to heterogeneous domestic AI accelerators (Ascend NPU, DCU, MLU, MUSA); resolved 30 of 47 triaged engineering issues, redesigned task-liveness detection, and built a unified result-integrity validator that flagged all 99 historical faulty records with zero false positives in replay.
- Root-caused a WER degradation of Parakeet-CTC-1.1B on Ascend NPU via layer-wise operator comparison (1,811 ops): an NPU-only branch in HuggingFace Transformers cast the float relative-position-bias mask to boolean, discarding positional information across 42 attention layers; fix verified on a 200-sample regression and reported upstream.
- Led a cross-platform accuracy-drift audit against GPU baselines, separating engineering artifacts from genuine platform drift across 8 outlier models and establishing comparability pre-checks (model ID, weights, inference engine, decoding parameters) for chip-accuracy statistics.

Shenzhen Quant-Cloud Energy Network Technology Co., Ltd.

Aug 2024

R&D Intern, Artificial Intelligence Department Shenzhen, China

- Built a bearing-temperature anomaly detector for wind-turbine gearboxes using Anomaly Transformer on historical sensor data for predictive maintenance; optimized preprocessing to improve detection accuracy.

Kenexs Intelligent Technology Co., Ltd.

Jul 2023 – Sep 2023

Software Development Intern Shenzhen, China

- Built a Qt-based log analyzer to visualize and filter logs from industrial machine-vision cameras, improving testers' debugging efficiency; fixed critical bugs and refined the interface for robustness.

SKILLS

Programming: Python, C/C++, CUDA, Java, SQL, JavaScript

ML & Systems: PyTorch, HuggingFace Transformers, SGLang, NeMo, NumPy/SciPy, CUDA kernel optimization (Tensor Cores, cuBLAS), Linux, Git, LaTeX

Languages: Chinese (native), English (TOEFL iBT 94, Nov 2024)